

Ajeet Krishnasamy

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Mechanical engineering graduate specializing in aerodynamics and computational fluid dynamics with experience in Python-driven simulation, optimization, and control systems.

EXPERIENCE

Systems Engineer, Medical Equipment (Intern)

Oct 2020 – Sept 2021

CenTrak

Newtown, PA, United States

- Engineered automated reliability analysis pipelines for medical device telemetry, processing real-time performance data from 10,000+ RTLS tags to quantify failure rates and improve system monitoring precision.
- Developed automated performance dashboards for the Connect Pulse™ RTLS platform, enabling engineering teams to identify device degradation trends and reduce unplanned maintenance events.
- Conducted battery discharge and device health analysis across 5,000+ medical-grade IoT devices, identifying failure precursors and enabling predictive replacement scheduling that prevented equipment outages.
- Standardized structured query and analysis workflows for multi-terabyte RTLS datasets, reducing troubleshooting cycle time through automated anomaly detection and root-cause analysis scripts.

Systems Engineer, Medical Equipment (Intern)

July 2019 – Sept 2019

CenTrak

Newtown, PA, United States

- Performed system-level characterization of RTLS localization algorithms using statistical regression analysis, identifying coverage gaps and refining location accuracy through receiver parameter tuning.
- Designed and executed validation test protocols for RTLS hardware deployment, correlating laboratory calibration data with field performance metrics to close the lab-to-field validation gap.
- Synthesized system performance data into technical recommendations for hardware upgrades and feature rollouts, supporting engineering decisions with quantitative evidence.

PROJECTS

AK-Vortex: High-Performance C++ Lattice Boltzmann CFD Solver | *C++, Python, CFD*

- Built a custom 2D Lattice Boltzmann Method (D2Q9) CFD solver in C++20 from first principles, using a cache-optimized flat std::vector layout and collapsed(2) OpenMP collision and streaming for multi-core scaling.
- Implemented a Multi-Relaxation-Time collision operator, Bouzidi interpolated bounce-back for curved boundaries, and a Smagorinsky LES subgrid model, validating 15 cases against canonical benchmarks (Blasius, Ghia, ERCOFTAC 043, Kutta-Joukowski) backed by a Google Test suite and CI.
- Currently extending the solver with physics-informed neural networks (PINNs) to accelerate and supervise solutions across multiple flow cases.

SwarmGNC: Decentralized UAV Formation Control | *Python, GNC*

- Built a decentralized multi-agent UAV swarm simulation with 6-DOF state-space dynamics (RK4 at 50 Hz), LQR optimal control via CARE solver, and 3D Direction Cosine Matrix heading-aligned guidance for a 7-drone wedge formation.
- Implemented Artificial Potential Field obstacle avoidance with 3D signed-distance sphere repulsion, graph Laplacian consensus for formation keeping, and Dryden wind turbulence modeling (Gauss-Markov, $\sigma = 2$ m/s).
- Conducted Monte Carlo parameter sensitivity analysis across 10 gain values, identifying four behavioral regimes and failure modes at extreme parameter combinations.

Airfoil CFD Explorer: Automated Aerodynamic Analysis & Shape Optimization | *Python, CFD (SU2)*

- Built an automated CFD pipeline solving compressible RANS on hybrid C-grid meshes across 5 angles of attack ($0^\circ - 16^\circ$), generating Cl/Cd polars with convergence history for each regime.
- Validated NACA 0012 aerodynamic coefficients against Ladson 1988 wind tunnel data ($Re = 1 \times 10^6$, $M = 0.15$), benchmarking SU2 RANS predictions against experimental lift and drag polars.
- Optimized an airfoil at $\alpha = 4^\circ$ via CST parameterization (16 Bernstein weights), NeuralFoil, and SciPy SLSQP; minimizing drag at target lift under geometric constraints, with SU2 RANS verification.

TECHNICAL SKILLS

Aerodynamics & CFD: SU2, ANSYS CFD, PhiFlow, OpenFOAM

Programming & CAD/FEA: Python (NumPy, Pandas, SciPy, Matplotlib), MATLAB, C++, SolidWorks, Fusion 360

GNC: LQR Optimal Control, APF Guidance, Graph Laplacian Consensus, DCM, Monte Carlo Analysis

Manufacturing: CNC Machining, 3D Printing, Lathe & Mill Operations, MIG Welding

EDUCATION

University of Ottawa

Ottawa, ON, Canada

Bachelor of Applied Science (BASc), Biomedical Mechanical Engineering

May 2026

Coursework: Fluid Mechanics, Heat Transfer, Control Systems, Vibration Analysis, Machine Design, CAD/FEA

Accreditation: CEAB-accredited; ABET-equivalent